Welcome to the MINI Dynamic Planet Division C Test

1. (3.00 pts) This occurs in a braided stream:

(Mark **ALL** correct answers)

- [ ] A) Increased river velocity
- [x] B) Bar formation
- [x] C) Deposition
- [x] D) Sorting of sediment

2. (4.00 pts) What is a difference between a delta and an alluvial fan?

(Mark **ALL** correct answers)

- [ ] A) A delta is formed when a river emerges from a mountainous region and flattens suddenly, causing the sediment to be deposited, while alluvial fans form when the river velocity slows at the mouth of the river and sediment is deposited
- [x] B) A delta is formed when the river velocity slows at the mouth of the river and sediment is deposited, while alluvial fans form when a river emerges from a mountainous region and flattens suddenly, causing the sediment to be deposited.
- [ ] C) A delta is an erosional feature, whereas an alluvial fan is a depositional feature.
- [ ] D) A delta is a depositional feature, whereas an alluvial fan is an erosional feature.

3. (4.00 pts) .In which of the following locations are karst features found?

(Mark **ALL** correct answers)

- [x] A) Pennsylvania
- [x] B) Kentucky
- [x] C) Florida
- [x] D) Alaska

4. (4.00 pts) What is the difference between a parallel drainage pattern and dendritic drainage pattern?

(Mark **ALL** correct answers)

- [x] A) . Parallel is sometimes indicative of an underlying fault
- [ ] B) Dendritic indicates that there are rock bands of different densities (aka, not homogeneous) underneath the river
- [ ] C) Dendritic is found where there is a discernable slope to the surface
- [ ] D) Parallel tends to be more common

5. (4.00 pts) Which of the following tools can be used to measure turbidity of water?

(Mark **ALL** correct answers)

- [x] A) Secchi Disk

- [ ] B) Peterson Grab
- [ ] C) Nansen Bottle
- [x] D) Nephelometer

6. (4.00 pts) The Continental Divide/Great Divide is:

(Mark **ALL** correct answers)

- [ ] A) A boundary that separates climate types.
- [x] B) A boundary that shows the separation of water flowing into the Atlantic Ocean from water flowing into the Pacific.
- [ ] C) A boundary that separates two continents.
- [x] D) A boundary that separates two drainage basins.

7. (4.00 pts) What is formed when a meander is cut off from the rest of the river?

(Mark **ALL** correct answers)

- [ ] A) Bar
- [x] B) Oxbow Lake
- [x] C) Billabong
- [ ] D) Isolated Lake

8. (4.00 pts) All of the following are examples of point-source contaminants except for:

(Mark **ALL** correct answers)

- [ ] A) Industrial facilities dumping waste
- [x] B) Bacteria and nutrients from livestock, and pet wastes
- [ ] C) Drainage outlet delivering polluted runoff
- [x] D) Fertilizers, herbicides and insecticides from agricultural lands and residential areas

9. (4.00 pts) What is a major result of urban development?

(Mark **ALL** correct answers)

- [ ] A) Less runoff
- [x] B) More impervious surface
- [x] C) Flooding
- [x] D) More runoff

10. (1.00 pts) What is the relationship between groundwater and surface water?

- (●) A) Surface water percolates through the soil to form groundwater
- ( ) B) There is no relationship between surface and groundwater, they form independently
- ( ) C) Groundwater filters through the soil to form surface water.
- ( ) D) Surface water is found in lakes directly above underground water sources

11. (1.00 pts) Estuaries _____

- A) are always saltwater ecosystems
- B) are always freshwater ecosystems.
- C) are ecosystems where both fresh water and salt water are present.
- D) prevent the development of salt marshes.

12. (1.00 pts) The largest watershed in the United States is the

- A) Mississippi River
- B) Yukon River.
- C) Colorado River.
- D) Rio Grande.

13. (1.00 pts) An estuary functions as a mineral trap due to

- A) its great depth and particularly sticky sediment
- B) the mixing action of river water and ocean water.
- C) its storing of pesticides from agricultural runoff.
- D) the filtering action of bottom-feeding fish species.

14. (1.00 pts) Parasitic worms would be classified as which type of water pollutant?

- A) pathogen
- B) organic matter
- C) inorganic chemicals
- D) heavy metals

15. (1.00 pts) As they flow down a mountain to flatter ground, rivers generally become

- A) wider.
- B) warmer.
- C) slower.
- D) All of the above

16. (1.00 pts) The types of organisms found in a pond or lake depend on

- A) the amount of sunlight available.
- B) the temperature of the water.
- C) the presence of nutrients.
- D) All of the above

17. (1.00 pts) Most of Earth's fresh water is

- A) suspended in atmospheric cloud formations
- B) held in reservoirs behind huge dams.
- C) solidified at the North and South Poles.
- D) stored in large underground rock structures

18. (1.00 pts) Swamps are commonly found on

- A) rocky hillsides.
- B) flat, poorly drained land
- C) forest clearings
- D) bottoms of ponds or lakes

19. (1.00 pts) The geological layer, consisting of underground caverns and porous layers of sand, gravel, or bedrock, where groundwater flows, is called

- A) the zone of saturation
- B) the water table
- C) an aquifer
- D) surface water
- E) the bedrock

20. (4.00 pts) Which of the following terms are associated with erosion?

(Mark **ALL** correct answers)

- [ ] A) Flocculation
- [x] B) Hydraulic Action
- [x] C) Corrasion
- [ ] D) Saltation
- [x] E) Corrosion
- [x] F) Attrition

21. (1.00 pts) What mainly distinguishes nekton from benthos?

- A) One lives in a freshwater ecosystem and the other lives in salty water.
- B) One swims freely and the other often lives attached to a hard surface
- C) One is a predator and the other is a primary producer
- D) All of the above

22. (1.00 pts)
If frozen ground can act as a low-conductivity aquitard, then the unfrozen ground beneath it can serve as a ____ ____, and the unfrozen active layer above it can be considered as a _____ _____

- A) Conned Aquitard; Conned Aquifer
- B) Unconned Aquitard; Conned Aquifer

C) Unconned Aquifer; Conned Aquifer

D) Perched Aquifer; Unconned Aquifer

E) Conned Aquifer; Perched Aquifer

23. (2.00 pts) The part of the stream discharge that is not attributable to direct runoff from precipitation or melting snow is known as _____ _____.

base flow

24. (2.00 pts) _____ is the process by which soluble materials in the soil, such as salts, nutrients, pesticide chemicals, or contaminants, are washed into a lower layer of soil or are dissolved and carried away by water

leaching

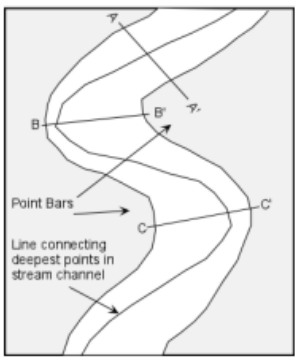
25. (2.00 pts) The vertical distance between the water level in a well before pumping and the water level in the well during pumping is known as _____.

drawdown

26. (2.00 pts) The three types of load carried by a river are suspended load, dissolved load, and _____ load.

bed

27. (4.00 pts)



1. The line connecting the deepest points in the stream channel is known as the _____.

2. This is a diagram of a(n) _____ channel.

thalweg

meandering

28. (2.00 pts) The imaginary line which divides the saturated zone and unsaturated zones: _____ _____.

water table

29. (2.00 pts) The _____ of ice water causes rock disintegration.

refreezing

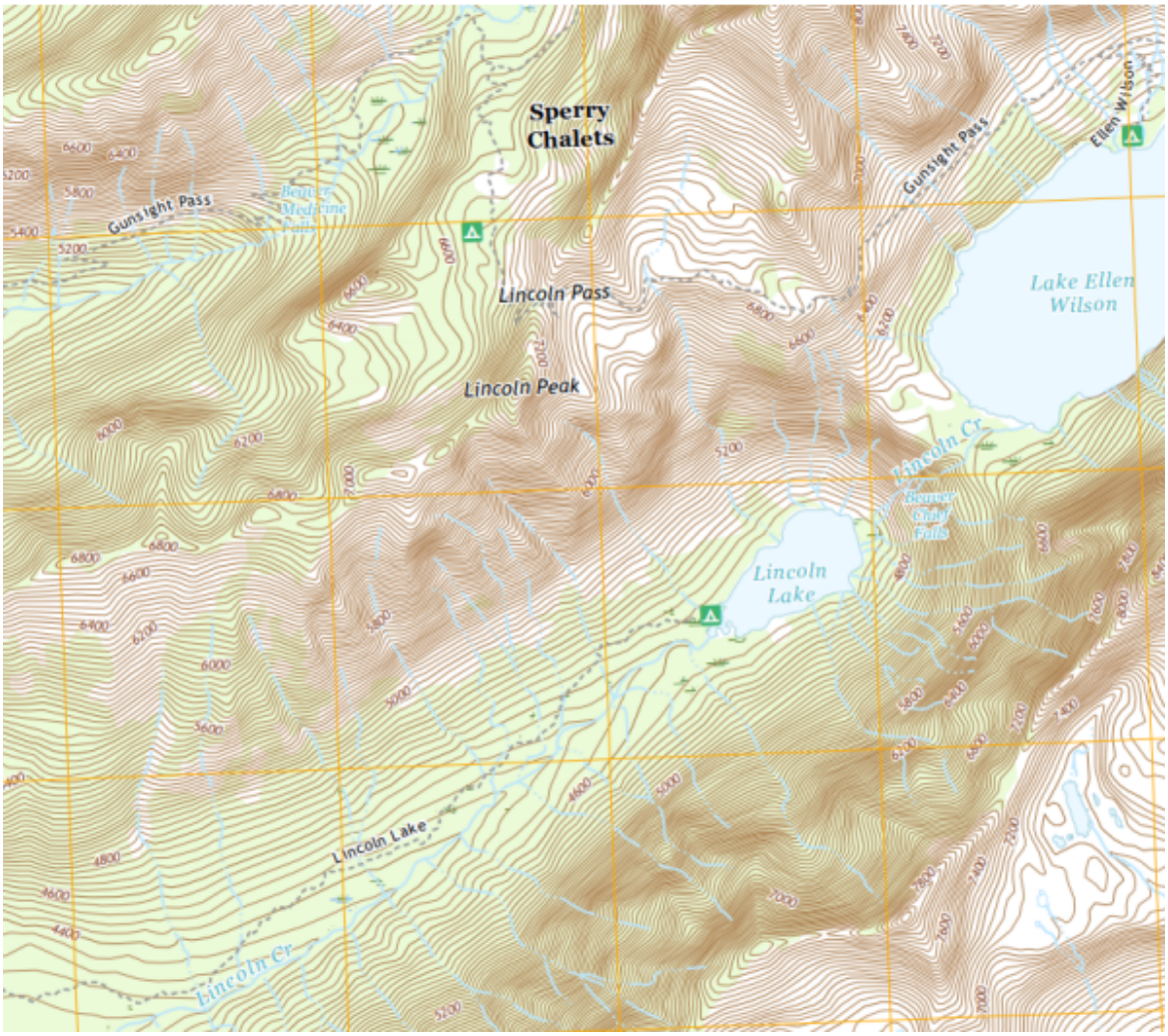
30. (2.00 pts) _____ valleys are the most common kinds of valley on the globe.

flat-floored

31. (2.00 pts) Some valleys sink below sea level. When they do, they are called _____ valleys.

sunken

Figure A: Map of Lincoln Lake and surrounding area. Elevation in feet.



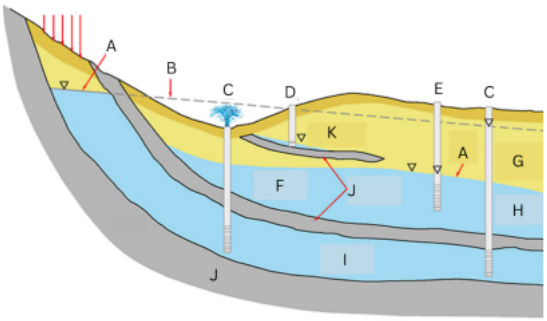
32. (6.00 pts) _____ erosion is the most likely type of erosion responsible for the specific erosional patterns shown in the map.

A _____ lake would best describe Lincoln Lake?

The stream southwest of Lincoln Lake, named Lincoln Creek, is heading in what direction?

glacial | Paternoster | southwest

33. (4.00 pts)

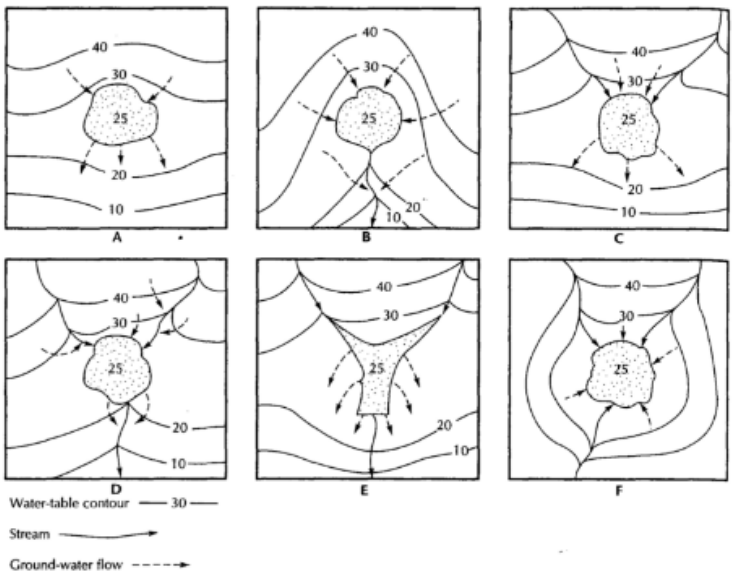


A. Which letter refers to a feature that is common at the base of slopes in hilly terrain? _____

B. What is its name? _____ _____

C | Artesian well

34. (6.00 pts)



The lake represented in stream F is a _____ lake. What is the primary way that water is discharged? _____

In the middle of Whoville is the lake represented by the letter A. Concerned about the bleak scenery, Carrie plants willow trees all around the border of the lake. The water table would be _____ by Carrie's actions?

terminal | evaporation | depressed

35. (2.00 pts) Hawaii is made up of many different volcanoes that rose up out of the ocean and made land. A common form of these volcanoes is a large central peak with very slowly sloping slopes radiating in all directions. What type of drainage pattern would you expect to see on such a mountain?

_____ drainage

radial

36. (2.00 pts) Sometimes, due to the effect in the previous question, one stream may erode back into the bed of another stream into which it gets diverted. The name of this process is stream _____.

capture or piracy

37. (4.00 pts) _____ and _____ are the two factors that make up evapotranspiration? (ENTER ALPHABETICALLY)

,

Evaporation

transpiration

38. (4.00 pts) _____ deals with water entering the ground (breaking the surface), _____ involves water passing through any rock or porous medium.

Infiltration

percolation

39. (2.00 pts) A block of ice falls off a glacier as the glacier retreats it then melts and creates a _____ lake.

kettle

40. (2.00 pts) Lake Tahoe has many rivers that feed into it yet only one that flows out of the lake and the water in the lake never reaches the ocean. This classifies the Tahoe river and lake system as what?

Endorheic

41. (2.00 pts) Order the 4 sediment types in the Wentworth size class based on increasing size.

,A: gravel , B: Mud, C: sand, D: silt (TYPE LETTER ORDER NO SPACES, NO COMMAS)

BDCA

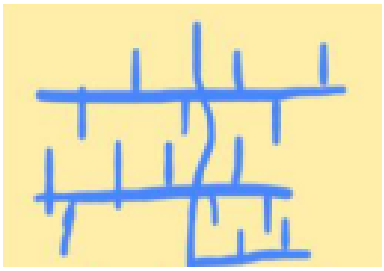
42. (2.00 pts) _____ zone is another name for the saturated zone?

Phreatic

43. (2.00 pts) A _____ _____ is A hole in the ground that allows you to see an underground river.

Karst fenster

44. (2.00 pts)



This is an diagram of a _____ drainage system

Trellis

45. (2.00 pts) What do you call the point where a tributary meets the mainstem?

Confluence

46. (12.00 pts) Describe the formation of a waterfall.

Expected Answer: When the river courses over resistant bedrock, erosion happens slowly (3 point), while downstream the erosion occurs more rapidly (3 point; if they explain both this and the previous point, add a point). As the watercourse increases its velocity at the edge of the waterfall, it plucks material from the riverbed (3 point). The water carries away th eroded sediment (3 point).

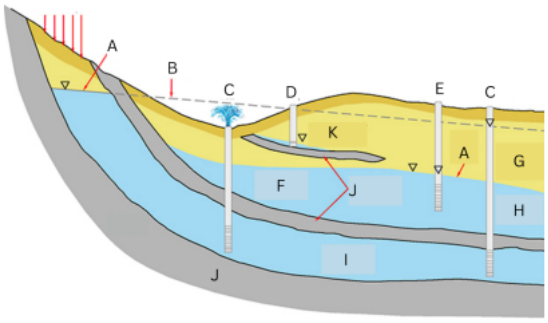
47. (10.00 pts) Describe a perched aquifer. Try to toss in the key terms.

Expected Answer: An aquifer which lays above the general water table line. There is an aquiclude/impermeable layer underlying the aquifer.

48. (9.00 pts) List 3 factors that influence infiltration rate.

Expected Answer: Soil type, vegetation, slope of the ground, current soil moisture, other reasonable answers.

49. (10.00 pts)



Why is location of point A significant in identifying effluent or influent streams?

Expected Answer: Effluent streams gain water from the saturated zone, while influent streams lose water to the saturated zone. When the water table is high, the saturated zone is higher, so streams are more likely to gain water and vice versa.

50. (10.00 pts)
The Colorado river discharges about 7.128 x 10^11 cubic feet per year. Say that an average 1st order stream discharges 1 cubic m/s (made up number don't put this in your binder lol).

What would be the Shreve order of the Colorado river?

(1 meter = 3.28 feet)

Expected Answer: 641 (accept 640 and 642). 7.128 x 10^11 ft^3 x (1 m/3.28 ft)^3 = 2.019 x 10^10 m^3/yrx 1 year/365 days x 1 day/24 hours x 1 hour/60 minutes x 1 minute/60 seconds = 640.53 m/s divided by 1 m/s = 641

CONGRATULATIONS you have reached the END of the TEST

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